

MedAxis Intelligence — Responsible AI Framework (Starter Template)

This document provides the MedAxis Responsible AI Framework template. You are required to complete and extend this framework as part of Phase 11 of the simulation. The sections marked [COMPLETE IN PHASE 11] require your analysis and documentation.

1. Model Governance Principles

MedAxis follows the NHS AI Lab's 'AI and Data in Health and Care Action Plan' and the EU AI Act's requirements for high-risk AI systems in healthcare. All models deployed in clinical pathways must meet the following principles:

- 1. Clinical Safety — models must not increase risk of harm to patients
- 2. Transparency — predictions must be explainable to clinicians
- 3. Fairness — models must be audited for performance disparities across protected characteristics
- 4. Accountability — every prediction must be logged with model version and timestamp
- 5. Human Oversight — clinical staff must retain authority to override any model output
- 6. Data Minimisation — only the minimum necessary data is used for each prediction

2. Model Card Template

2.1 Model Overview

Field	Value
Model Name	[COMPLETE IN PHASE 5]
Model Version	[COMPLETE IN PHASE 5]
Model Type	[COMPLETE IN PHASE 5]
Intended Use	[COMPLETE IN PHASE 5]
Primary Users	[COMPLETE IN PHASE 5]
Out-of-Scope Uses	[COMPLETE IN PHASE 5]
Regulatory Classification	[COMPLETE IN PHASE 5]
Training Data Period	[COMPLETE IN PHASE 5]
Feature Count	[COMPLETE IN PHASE 5]
AUC (test set)	[COMPLETE IN PHASE 4]
Recall at selected threshold	[COMPLETE IN PHASE 5]
Recommended Decision Threshold	[COMPLETE IN PHASE 5]

2.2 Fairness Audit Requirements

You must evaluate model performance across the following subgroups and document results in Phase 5:

Subgroup Variable	Groups to Test	Metric to Report
Age Band	18–40, 41–60, 61–75, 76+	AUC, Recall, False Negative Rate
Gender	Male, Female, Unknown	AUC, Recall, False Negative Rate
Ethnicity	All groups with n>50	AUC, Recall, Calibration
Deprivation Decile	Deciles 1–3 vs 8–10	Recall, False Negative Rate

Subgroup Variable	Groups to Test	Metric to Report
Hospital Site	MER001–MER006	AUC, Calibration
Admission Type	EM vs EL	Recall, Precision

3. Bias Mitigation Strategies

[COMPLETE IN PHASE 5] — Document the bias mitigation strategies you applied and why. Options include: threshold adjustment by subgroup, re-sampling, re-weighting, feature removal, separate models per subgroup.

4. Regulatory Compliance Checklist

Requirement	Framework	Status
Model registered as SaMD before clinical deployment	MHRA / EU AI Act	[PHASE 11]
DTAC assessment completed	NHS DTAC	[PHASE 11]
DPIA (Data Protection Impact Assessment) submitted	UK GDPR	[PHASE 11]
Technical documentation package prepared	EU AI Act Art. 11	[PHASE 11]
Human oversight mechanism documented	EU AI Act Art. 14	[PHASE 11]
Audit trail for all predictions enabled	EU AI Act Art. 12	[PHASE 10]
PHI de-identified in training data	HIPAA	[PHASE 2]
Model card published to internal registry	NHS AI Lab standard	[PHASE 5]

5. Drift Monitoring Plan

[COMPLETE IN PHASE 11] — Document your drift monitoring strategy. Must include: data drift thresholds, prediction drift thresholds, retraining triggers, monitoring frequency, alerting mechanism.

6. Incident Response

If the model produces a false negative that contributes to a clinical incident, the following steps must be taken within 24 hours:

1. Model immediately suspended from active use
2. Incident logged in the MedAxis AI Safety Register
3. MedAxis Chief Medical Safety Officer notified
4. Meridian CMO notified within 4 hours
5. Root cause analysis completed within 5 business days
6. MHRA notified if incident meets serious adverse incident threshold